

Long-term effects of calorie or protein restriction on serum IGF-1 and IGFBP-3 concentration in humans

Summary Reduced function mutations in the insulin/IGF-I signaling pathway increase maximal lifespan and health span in many species. Calorie restriction (CR) decreases serum IGF-1 concentration by ~40%, protects against cancer and slows aging in rodents. However, the long-term effects of CR with adequate nutrition on circulating IGF-1 levels in humans are unknown. Here we report data from two long-term CR studies (1 and 6 years) showing that severe CR without malnutrition did not change IGF-1 and IGF-1 : IGFBP-3 ratio levels in humans. In contrast, total and free IGF-1 concentrations were significantly lower in moderately protein-restricted individuals. Reducing protein intake from an average of 1.67 g kg⁻¹ of body weight per day to 0.95 g kg⁻¹ of body weight per day for 3 weeks in six volunteers practicing CR resulted in a reduction in serum IGF-1 from 194 ng mL⁻¹ to 152 ng mL⁻¹. These findings demonstrate that, unlike in rodents, long-term severe CR does not reduce serum IGF-1 concentration and IGF-1 : IGFBP-3 ratio in humans. In addition, our data provide evidence that protein intake is a key determinant of circulating IGF-1 levels in humans, and suggest that reduced protein intake may become an important component of anticancer and anti-aging dietary interventions.